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Article

College Students' Distractions from Learning Caused by Multitasking in Online vs. Face-to-Face Classes: A Case Study at a Public University in Romania

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Abstract: This study compared Romanian college students' off-task use of digital devices in online courses with their digital distractions in face-to-face courses. Students taking both online and face-to-face courses completed surveys assessing off-task use of technology in each delivery mode, extent to which such use constitutes a distraction, and instructor policies for curbing use of digital devices in class. Off-task use of digital devices was significantly greater in online than face-to-face courses. Students were twice as likely to state that they were distracted by multitasking in online than in face-to-face classes. They also indicated that instructors in online classes were less likely than their face-to-face instructors to have a policy on the use of digital devices for off-task purposes during class. Study participants were more lenient toward the use of digital devices in online than in in-person classes and were less bothered by multitasking and classmates' off-task use of digital devices in online than in face-to-face classes. The results of this study have implications for instructors in online classes. By improving instructional design and pedagogical methods and creating opportunities for more interaction during the class, online instructors might be able to increase student motivation and lower multitasking behaviors.

Keywords: multitasking; digital distractions; online education; face-to-face classes; impact on learning

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1. Introduction

Today's college students live in a complex world, a world full of distractions and interruptions caused by social media that may prevent them from achieving their educational goals. The term "interference," which includes distractions and interruptions, is commonly used to describe something that slows down, obstructs, impedes, and largely diverts another process. Interferences are generated both internally, in the form of thoughts that appear unexpectedly in our mind, and externally, due to sensory stimuli such as noise, beeps, vibrations, light signals produced by digital devices [1]. Distractions are fragments of information irrelevant to our tasks, which we either encounter outside or generate ourselves inside our mind, and interruptions occur when one makes the decision to engage in more than one task at a time. Interruptions in which people try to perform two or more tasks simultaneously aimed at goals that are independent of each other, are often referred to as multitasking [2]. College students are facing an explosion in the variety and accessibility of technologies with enticing sounds, pleasing visual effects and insistent vibrations that demand their attention, while their brains are battling multiple streams of information at the same time [2], thus affecting learning.

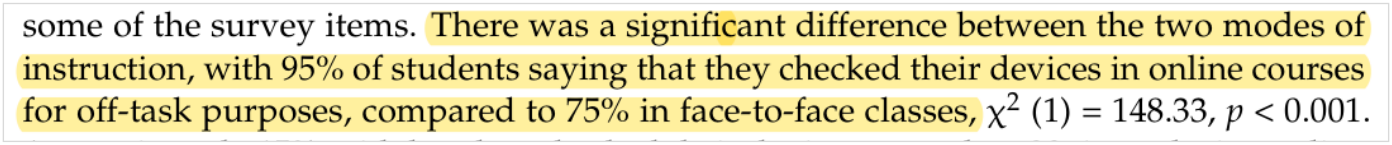
The goal of this paper is to add to the discussion of the impact of distractions on learning by examining how college students at a Romanian public university perceive the off-task (unrelated to school) use of technology in class. This behavior has often been referred to as "cyberslacking." Analyzing data from student surveys, we examine (1) how

How the article supports my concept

Aivaz & Teodorescu (2022). The images below are actual crops from the uploaded article; yellow highlights mark the passages I use.

1. More off-task device use in online lessons

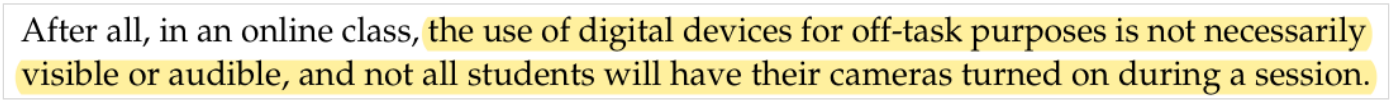
Results, original PDF p. 6


some of the survey items. There was a significant difference between the two modes of instruction, with 95% of students saying that they checked their devices in online courses for off-task purposes, compared to 75% in face-to-face classes, χ^2 (1) = 148.33, $p < 0.001$.

The authors report that 95% checked devices for off-task purposes in online classes, compared with 75% in face-to-face classes. This supports the problem I noticed during COVID: students could attend an online lesson while using other apps or phones.

2. Off-task activity can be difficult to notice

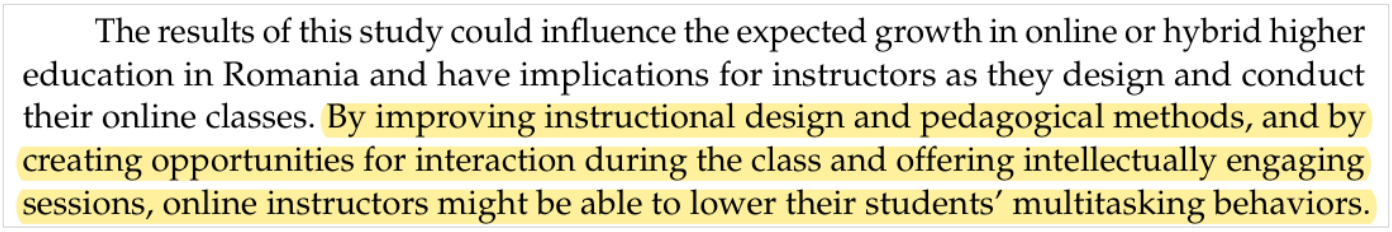
Discussion, original PDF p. 11


After all, in an online class, the use of digital devices for off-task purposes is not necessarily visible or audible, and not all students will have their cameras turned on during a session.

The authors explain that off-task activity may be neither visible nor audible online. This helps explain why looking present on Zoom does not always mean being engaged. It motivates my exploration of a classroom-like space with visible learning activities.

3. More interaction is a direction worth testing

Conclusions, original PDF p. 11


The results of this study could influence the expected growth in online or hybrid higher education in Romania and have implications for instructors as they design and conduct their online classes. By improving instructional design and pedagogical methods, and by creating opportunities for interaction during the class and offering intellectually engaging sessions, online instructors might be able to lower their students' multitasking behaviors.

The authors suggest that engaging lessons and more interaction might reduce multitasking. This supports testing my shared whiteboard, personal desk paper and controller-based writing as ways to take part actively in a lesson. Whether these VR interactions improve focus still needs testing.

Scope of the evidence. This was a self-report survey of 630 economics students at one Romanian university. It supports my problem statement and design rationale; it did not test VR or demonstrate better focus or grades. The abstract and discussion conflict on the direction of the "twice as distracted" claim, so I do not rely on that claim.

Source: Aivaz, K. A., & Teodorescu, D. (2022). *International Journal of Environmental Research and Public Health, 19*(18), 11188. https://doi.org/10.3390/ijerph191811188
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